

HYDROGENATED NBR (HNBR) PATENT LANDSCAPE: HYDROGENATION, METATHESIS-ENABLED MOLECULAR WEIGHT CONTROL, AND DISPERSION/BINDER APPLICATIONS

1. ABSTRACT

The patent set around hydrogenated nitrile butadiene rubber (HNBR) is dominated by two technical themes: first, synthesis or precursor modification routes that control the molecular weight and residual unsaturation of nitrile rubber before or during hydrogenation; and second, downstream formulations that exploit HNBR as a dispersant, binder, or interface modifier in conductive or rubber compositions. The core material definition is consistent across the HNBR-focused patents: HNBR is produced by hydrogenation of nitrile rubber, with the diene-derived C=C bonds partially or fully hydrogenated, and the hydrogenation degree commonly described as 50 to 100%. Several patents emphasize that commercial HNBR typically has relatively high Mooney viscosity and broad molecular-weight distribution, motivating approaches to lower viscosity while preserving processability and performance.

The synthesis-oriented patents (EP1825913B1 and US7737233B2) describe metathesis-based degradation of nitrile rubber, often followed by hydrogenation, to obtain lower-Mooney HNBR with narrower molecular-weight distribution. These documents highlight the use of ruthenium- or osmium-based metathesis catalysts, with salt additives in the newer catalyst-system patent, and note that co-olefins may be used in some routes. The reported target window for the degraded/hydrogenated products includes weight-average molecular weights of 30,000 to 250,000, Mooney viscosity of 3 to 50, and PDI below 2.5 in the cited process family. The newer catalyst-system patent also stresses that catalyst activity and reaction efficiency are critical, because prior approaches required relatively large catalyst loadings and long reaction times.

The formulation patents show a clear trend toward using HNBR as a functional dispersant rather than as a bulk elastomer. US11108050B2 and US11658303B2 use hydrogenated nitrile-based rubber or related nitrile copolymers to stabilize conductive material dispersions for battery electrodes, with control of viscoelastic properties such as complex modulus, phase angle, and Mooney viscosity. EP3348582B1 and EP3355392B1 specifically use partially hydrogenated nitrile rubber to disperse bundle-type or entangled-type carbon nanotubes, with residual double bond (RDB) values of 0.5 to 40 wt% and particle-size targets for the dispersed CNT aggregates.

Overall, the landscape shows a progression from synthesis/process control of HNBR precursors toward highly engineered dispersion and electrode-binder applications that depend on tuned residual unsaturation, viscosity, and interfacial affinity.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US11108050B2
- Patent Title: Conductive material dispersed liquid and secondary battery manufactured using ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 2

- Patent Number: EP1825913B1
- Patent Title: New catalyst systems and their application for metathesis reactions
- Assignee: Not disclosed

- Jurisdiction: EP
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 3

- Patent Number: US7737233B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 4

- Patent Number: US11658303B2
- Patent Title: Conductive material dispersion, binder resin-containing conductive material ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 5

- Patent Number: EP3348582B1
- Patent Title: Carbon nanotube dispersion liquid and manufacturing method thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 6

- Patent Number: EP3355392B1
- Patent Title: Carbon nanotube dispersion liquid and manufacturing method thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: EP3920285B1
- Patent Title: Binder composition for secondary battery electrode, conductive material paste ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 8

- Patent Number: US9976027B2
- Patent Title: Polymer fine particle-containing curable resin composition having improved ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 9

- Polymerization / Synthesis Method** - No polymerization parameters disclosed.

Technical Relevance Selected via deterministic pipeline scoring.

- Patent Number: US10815320B2
- Patent Title: Functionalized resin having a polar linker
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP1825913B1 and US7737233B2 both address metathesis-based degradation of nitrile rubber followed by hydrogenation to make lower-Mooney HNBR, but EP1825913B1 is broader in catalyst-system scope while US7737233B2 explicitly adds alkaline-earth metal chloride salts to the metathesis catalyst.
- Both metathesis patents describe nitrile rubber as a copolymer/terpolymer of at least one α,β -unsaturated nitrile and at least one conjugated diene, with optional additional copolymerizable monomers; the HNBR product is obtained by hydrogenating the diene-derived double bonds.
- EP1825913B1 states that HNBR hydrogenation degree is usually 50 to 100% and discusses low-Mooney HNBR targets below 55 Mooney or M_n below about 200,000 g/mol; it also cites metathesis/hydrogenation routes giving M_w 30,000 to 250,000, Mooney 3 to 50, and PDI below 2.5.
- US7737233B2 emphasizes that prior metathesis routes required relatively large catalyst amounts and long reaction times, and that the salt-additive catalyst system improves activity; it specifically mentions calcium chloride or magnesium chloride as preferred salts.
- EP3348582B1 and EP3355392B1 both use partially hydrogenated nitrile rubber as a dispersant for carbon nanotubes, but EP3348582B1 is directed to bundle-type CNTs with D50 of 3 to 10 μm , whereas EP3355392B1 is directed to entangled-type CNTs with D50 of 3 to 5 μm .
- The two CNT-dispersion patents share the same residual double bond window for the partially hydrogenated nitrile rubber: 0.5 wt% to 40 wt%, calculated from BD and HBD units; both also describe the rubber as containing α,β -unsaturated nitrile-derived, conjugated diene-derived, and hydrogenated conjugated diene-derived units.
- EP3348582B1 gives a broader CNT particle-size envelope, including D10 of 1 to 3 μm and D90 up to 30 μm , while EP3355392B1 gives a narrower entangled-CNT envelope, including D10 of 0.5 to 1.5 μm and D90 of 6 to 10 μm .
- Both CNT patents state that the partially hydrogenated nitrile rubber may be prepared by copolymerizing α,β -unsaturated nitrile, conjugated diene, and optionally other copolymerizable monomers, followed by hydrogenation of the C=C bonds; they also note that common polymerization and hydrogenation methods may be used.

- US11108050B2 uses a hydrogenated nitrile-based rubber as a dispersant in a conductive material dispersed liquid for battery electrodes, but the claim language is focused on rheology rather than HNBR synthesis; it specifies a complex modulus of 20 to 500 Pa at 1 Hz and a hydrogenated conjugated diene-derived repeating unit content of 1 to 15 wt%.
- US11658303B2 similarly uses a nitrile-containing copolymer as a dispersant in conductive material dispersion, but it is not an HNBR synthesis patent; it specifies a copolymer A with 15 to 50 wt% nitrile-group structural units, 40 to 85 wt% aliphatic hydrocarbon structural units, and Mooney viscosity of 40 to 70.
- EP3920285B1 is also formulation-oriented rather than synthesis-oriented: it uses a copolymer with 40 to 70 mass% alkylene units and 30 to 55 mass% nitrile-group monomer units, Mooney viscosity of 25 or less, and iodine value of 8 mg/100 mg or more to improve conductive material dispersion in battery electrodes.
- US9976027B2 and US10815320B2 are only indirectly related to HNBR; they discuss diene-rubber core-shell particles and silane-functionalized resins, respectively, for rubber or adhesive formulations, but they do not provide HNBR synthesis parameters and are not primary HNBR process disclosures.
- Across the set, the recurring technical strategy is to tune residual unsaturation, molecular weight, and viscoelasticity to balance processability with interfacial performance; the newer application patents shift from bulk rubber properties toward dispersion stability, CNT wetting, and electrode performance.
- A historical-to-newer trend is visible: older patents focus on metathesis degradation plus hydrogenation to lower Mooney viscosity, while newer patents focus on using partially hydrogenated nitrile rubber as a tailored dispersant in high-performance conductive formulations.

4. CONCLUSION

The evidence indicates that the most robust HNBR synthesis route in this patent set is precursor molecular-weight reduction by metathesis of nitrile rubber, followed by hydrogenation, using ruthenium- or osmium-based catalysts; salt-assisted catalyst systems are presented as a later improvement to raise metathesis activity and reduce the burden of catalyst loading and reaction time. For HNBR product targets, the patents consistently favor moderate residual unsaturation in the precursor or partially hydrogenated product, with hydrogenation degrees commonly described as 50 to 100% and, in the CNT-dispersion patents, residual double bond values of 0.5 to 40 wt%. For processability, the most relevant low-viscosity HNBR targets are Mooney viscosities below 55, and in the metathesis-derived examples, as low as 3 to 50, with narrower molecular-weight distributions preferred. For formulation use, the best-supported parameters are controlled viscoelasticity and particle-size management: hydrogenated nitrile-based dispersants with low-to-moderate modulus, and partially hydrogenated nitrile rubber that matches CNT aggregate size and surface wetting requirements. Overall, the patent landscape supports HNBR synthesis strategies that first control chain length and unsaturation, then exploit that tuned chemistry in dispersion-stable, battery-relevant formulations.

5. REFERENCES

- US11108050B2 | Conductive material dispersed liquid and secondary battery manufactured using ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11108050B2/en>
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- US10815320B2 | Functionalized resin having a polar linker | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10815320B2/en>